

Rain Detector System Using Arduino

PROJECT FINAL REPORT

Name: P.G.R.H.Pusswella

Faculty of Engineering
Sri Lanka Technological Campus

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Contents

I. Abstract	3
II. Introduction	3
III. Objective.....	3
III. Problem Statement	4
IV. Components Used.....	4
V. Circuit Diagram	5
Explanation of the Circuit Diagram	5
1. Rain Sensor Connections	5
2. Servo Motor Connections	6
3. Buzzer Connections.....	6
4. Arduino Nano Power Connections.....	6
VI. Working Principle.....	6
VII. Arduino Code & Explanation	7
1. Code:.....	7
2. Code Explanation	8
VIII. Results & Final Outcome.....	9
1. Results	9
2. Final Outcome.....	9
IX. Real-Life Applications	9
X. Advantages of the System	10
XI. Future Improvements	10
XII. Conclusion	10
XIII. References	10

I. Abstract

This project presents the design and implementation of a rain detector system using an Arduino Nano. The system is intended to automatically detect rainfall and take appropriate actions, such as collecting clothes using a servo motor or triggering an alarm using a buzzer. The project utilizes a rain sensor, an Arduino Nano, a servo motor, a buzzer, and other essential components to create a functional and efficient rain detection system. The system is useful for household applications, agriculture, and automation systems. This project aims to develop a cost-effective and efficient solution for rain detection, providing an automated response to rainfall.

II. Introduction

Water conservation is crucial in today's world. Rainwater harvesting plays an essential role in storing and utilizing rainwater effectively. Various automation systems, including rain detection systems for irrigation, home automation, and automotive applications, can help in this process.

This project focuses on a rain detection-based automatic clothes collector system that protects clothes from getting wet. The rain sensor detects rainfall and triggers a servo motor to retract clothes while activating a buzzer to alert the user. The system ensures an automated and hassle-free response to rain, making it ideal for households, agricultural setups, and smart home applications.

III. Objective

The main objectives of this project are:

- To design a system that detects rain using a rain sensor and provides a real-time response.
- To automate clothes collection when it rains using a servo motor mechanism.
- To provide an alert system through a buzzer when rain is detected.
- To create a cost-effective, energy-efficient, and reliable rain detection system.
- To apply automation concepts to real-life problems such as rainwater harvesting, smart irrigation, and home automation.

III. Problem Statement

Unexpected rain can damage household items, especially clothes hung outside. Manually collecting clothes every time it rains is inconvenient. Additionally, excess water can harm crops in agriculture if not managed properly.

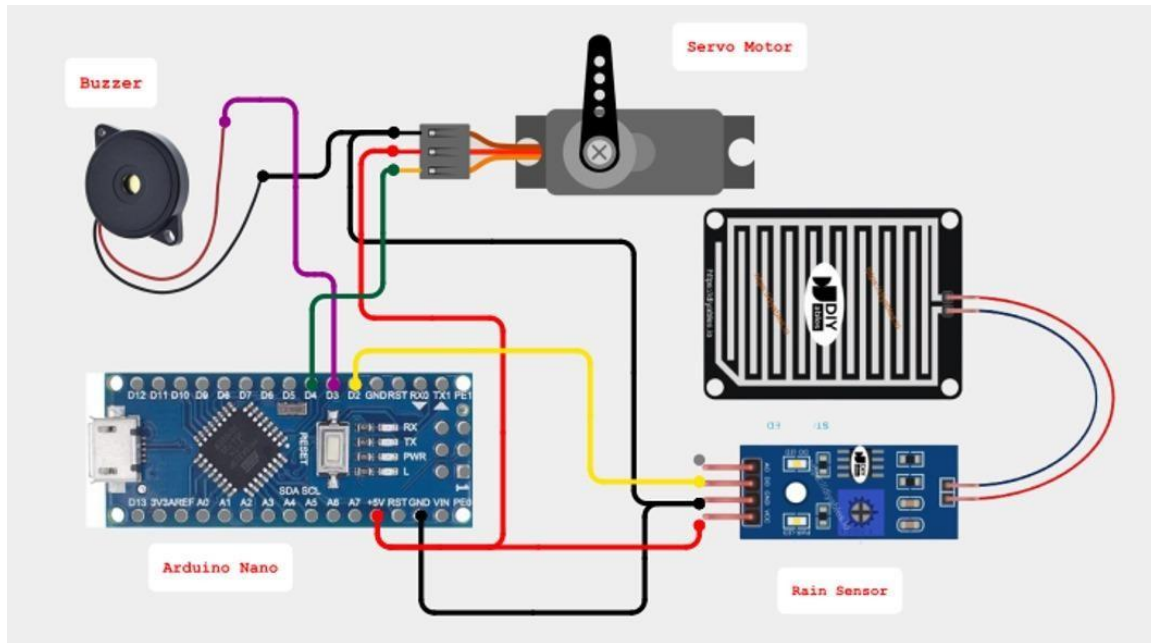
This project proposes an automatic rain detection system that:

- Detects rain immediately and activates an automatic response mechanism.
- Moves clothes to a safe position using a servo motor.
- Alerts users with a buzzer sound.
- Can be expanded for other applications like irrigation control, automobile rain sensors, and smart homes.

IV. Components Used

- Arduino Nano – The microcontroller responsible for processing inputs and controlling outputs.
- Arduino Nano Cable – Used for programming and power supply.
- Rain Drop Sensor – Detects rainfall and provides input signals.
- Servo Motor – Moves the clothes rack to a safe position.
- Buzzer – Emits an alert sound when rain is detected.
- Mini Breadboard – Facilitates connections between components.
- Jumper Wires (Male-to-Male & Male-to-Female) – Used for wiring connections.

V. Circuit Diagram



Explanation of the Circuit Diagram

This circuit connects an Arduino Nano to a rain sensor, servo motor, and buzzer to detect rain and perform automated actions. Here's how each component is connected:

1. Rain Sensor Connections

The rain sensor consists of two parts:

- Sensor Plate (detects rain)
- Control Module (processes signals) Connections:
- VCC (Red Wire) → Connected to 5V on Arduino
- GND (Black Wire) → Connected to GND on Arduino
- Analog Output (Yellow Wire) → Connected to D2 on Arduino

How it Works: When rainwater touches the sensor plate, it decreases resistance, sending a lower voltage signal to Arduino.

2. Servo Motor Connections

The servo motor is responsible for moving the clothes rack.

Connections:

- VCC (Red Wire) → Connected to 5V on Arduino
- GND (Brown Wire) → Connected to GND on Arduino
- Signal (Orange Wire) → Connected to D4 on Arduino

How it Works: When rain is detected, the Arduino sends a signal to move the servo motor to 90 degrees, retracting the clothes.

3. Buzzer Connections

The buzzer provides an audible alert when rain is detected.

Connections:

- Positive Terminal (Brown Wire) → Connected to D3 on Arduino
- Negative Terminal (Black Wire) → Connected to GND on Arduino

How it Works: When rain is detected, the Arduino sets D6 HIGH, activating the buzzer.

4. Arduino Nano Power Connections

- 5V Pin → Powers the rain sensor and servo motor
- GND Pin → Common ground for all components

VI. Working Principle

1. The system works based on the rain sensor's resistance principle:
2. The rain sensor detects water droplets on its surface.
3. The sensor output is processed by the Arduino Nano.
4. If rainfall is detected (sensor value drops below a set threshold):
 - The servo motor moves the clothes rack to safety.
 - The buzzer turns on to alert the user.
5. If no rain is detected, the system remains inactive.
6. The system continuously monitors rainfall and takes action when required.

VII. Arduino Code & Explanation

1. Code:

```
#include <Servo.h> // Pin Definitions const int
rainSensorPin = 2; // Rain sensor pin const int
buzzerPin = 3; // Buzzer pin const int servoPin
= 4; // Servo motor pin
Servo clothesServo; // Variables bool isRaining = false;
// Keeps track of rain status int rainValue = 0; //
Stores rain sensor value
// Function to move the servo slowly void moveServoSmoothly(int startAngle,
int endAngle, int stepDelay) { if (startAngle < endAngle) { for (int
angle = startAngle; angle <= endAngle; angle++) {
clothesServo.write(angle); delay(stepDelay); // Delay between each
step
} } else { for (int angle = startAngle; angle >= endAngle;
angle--) { clothesServo.write(angle); delay(stepDelay);
// Delay between each step
}

} } void setup() {
pinMode(rainSensorPin, INPUT);
pinMode(buzzerPin, OUTPUT);
clothesServo.attach(servoPin);
clothesServo.write(92); // Initial
position (drying position)

Serial.begin(9600); // For debugging

} void loop() { rainValue = digitalRead(rainSensorPin); if
(rainValue == LOW) { // LOW indicates rain detected if
(!isRaining) { isRaining = true;
digitalWrite(buzzerPin, HIGH); // Turn on the buzzer
moveServoSmoothly(92, 0, 15); // Move clothes to a covered
position slowly delay(1000); // Buzzer ON for 1
second digitalWrite(buzzerPin, LOW); // Turn off the buzzer
}

} else { // No rain detected
if (isRaining) { isRaining
= false;
moveServoSmoothly(0, 92, 15); // Move clothes back to drying position slowly
}

} delay(100); // Small delay to stabilize sensor
readings
}
```

2. Code Explanation

1. Library Inclusion:

- The Servo.h library is included to control the servo motor.

2. Pin Definitions:

- rainSensorPin (Pin 2) detects rain.
- buzzerPin (Pin 3) activates a buzzer when rain is detected.
- servoPin (Pin 4) controls the servo motor that moves the clothes cover.

3. Servo Object and Variables:

- clothesServo: Servo object to control the motor.
- isRaining: Boolean variable to track rain status.
- rainValue: Stores the rain sensor reading.

4. Function to Move Servo Smoothly:

- Moves the servo gradually between two angles.
- Uses a loop with a small delay to ensure smooth movement.
- Prevents sudden jerks when opening or closing.

5. Setup Function:

- Initializes the rain sensor and buzzer.
- Attaches the servo motor and sets it to the initial drying position (92°).
- Enables serial communication for debugging.

6. Loop Function:

- Continuously reads the rain sensor value.
- If rain is detected (LOW):
 - Checks if it was not already raining.
 - Turns ON the buzzer and moves the servo to 0° (covered position).
 - Keeps the buzzer ON for 1 second, then turns it OFF.
- If no rain (HIGH):
 - Checks if it was previously raining.
 - Moves the servo back to 92° (drying position).
- A small delay (100ms) prevents sensor noise and false triggers.

7. Final Behavior:

- The system automatically detects rain and moves the clothes to safety.
- Once the rain stops, the system returns them to the drying position.

VIII. Results & Final Outcome

1. Results

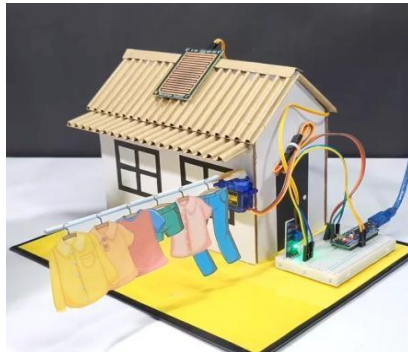
The rain detection system was successfully tested. The following results were observed:

- When the rain sensor detected water, the servo motor moved the clothes rack, and the buzzer activated.
- When the sensor was dry, the servo motor remained in its default position, and the buzzer stayed off.

2. Final Outcome

The system effectively:

- Detects rain using a rain sensor.
- Automatically moves clothes to a safe position.
- Alerts users through a buzzer.
- Can be integrated with other smart home applications.



IX. Real-Life Applications

This rain detection system can be extended to various real-world applications, including:

- Smart Homes – Automates clothes protection and window closure.
- Automobiles – Can trigger automatic windshield wipers.
- Agriculture – Helps in rainwater harvesting and irrigation control.
- Smart Cities – Used in weather monitoring and environmental studies.

X. Advantages of the System

- Fully Automated – No manual intervention needed.
- Low Cost – Uses affordable components.
- Energy Efficient – Consumes minimal power.
- Reliable & Fast – Provides instant response to rainfall.
- Expandable – Can be integrated with IoT and AI.

XI. Future Improvements

This project can be improved with additional features, including:

- IoT Integration – Sending real-time rain alerts to mobile devices.
- Weather Prediction AI – Using AI to predict rainfall in advance.
- Multiple Sensors – Covering a larger area for rain detection.
- Solar Power – Using solar panels as a sustainable energy source.

XII. Conclusion

The Rain Detector System using Arduino successfully detects rain and automates clothes protection. It provides an efficient, low-cost, and reliable solution for handling unexpected rainfall. The servo motor mechanism and buzzer alert system ensure quick action.

With future enhancements, such as IoT integration and AI-based weather predictions, the system can be further optimized for smart home automation, agriculture, and environmental monitoring.

This project demonstrates how Arduino-based automation can provide practical solutions to everyday problems.

XIII. References

- Arduino. (n.d.). *Servo Library Documentation*.
- Last Minute Engineers. (n.d.). *Interfacing Rain Sensor with Arduino*.
- ElectronicWings. (n.d.). *Buzzer Interfacing with Arduino Uno*.